

Name: Maths Class:

SYDNEY TECHNICAL HIGH SCHOOL



Year 11 Mathematics

Preliminary HSC Course

Yearly Exam

September, 2016

Time allowed: 2 hours

General Instructions:

- Marks for each question are indicated on the question.
- Approved calculators may be used
- All necessary working should be shown
- Full marks may not be awarded for careless work or illegible writing
- **Begin each question on a new page**
- Write using black or blue pen
- All answers are to be in the writing booklet provided

Section 1	Multiple Choice Questions 1-10 10 Marks
Section II	Questions 11-18 72 Marks
Total	82 marks

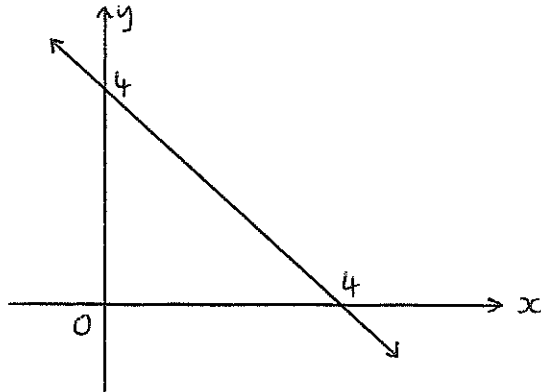
SECTION 1 -- MULTIPLE CHOICE (10 marks)

QUESTION 1

What is the gradient of a line parallel to the line $2x + 3y - 1 = 0$?

- A. 2 B. -2 C. $\frac{3}{2}$ D. $-\frac{2}{3}$

QUESTION 2



- The equation of the line above is: A. $x - y + 4 = 0$ B. $x + y - 4 = 0$
C. $x + y + 4 = 0$ D. $x - y - 4 = 0$

QUESTION 3

A function is given by $f(x) = \sqrt{9 - x^2}$. What is its natural domain?

- A. $x < 3$ B. $x \leq 3$ C. $-3 \leq x \leq 3$ D. $-9 \leq x \leq 9$

QUESTION 4

The function in Question 3 above is:

- A. even B. odd C. neither D. cannot be determined

QUESTION 5

What is the minimum value of $x^2 - 4x + 6$?

- A. 2 B. 4 C. 6 D. 8

QUESTION 6

If $a^b = 5$, what is the value of $2a^{3b}$?

- A. 30 B. 250 C. 500 D. 1000

QUESTION 7

If $3^{x-4} = 9^{2x}$, then $x = ?$

- A. $\frac{3}{4}$ B. $\frac{4}{3}$ C. $-\frac{3}{4}$ D. $-\frac{4}{3}$

QUESTION 8

If $2x^2 - 12x + 11$ is expressed in the form $2(x - b)^2 + c$, what is the value of c ?

- A. -25 B. -7 C. 2 D. 29

QUESTION 9

$\frac{\sin(180^\circ - \theta)}{\cos(90^\circ - \theta)}$ simplifies to:

- A. 1 B. 2 C. $\tan \theta$ D. $\cot \theta$

QUESTION 10

If $a > b$, which of the following is always true?

- A. $a^2 > b^2$ B. $\frac{1}{a} < \frac{1}{b}$ C. $-a > -b$ D. $2^a > 2^b$

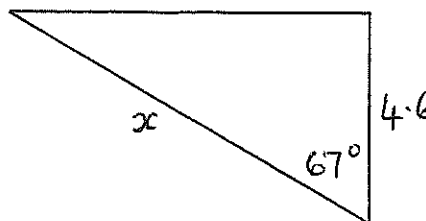
END OF SECTION 1

SECTION 2

QUESTION 11 (9 marks)

- a) Evaluate $13.6 \sin 42^\circ 15'$ correct to 2 significant figures. 1
- b) Expand and simplify $(2\sqrt{3} - 1)(\sqrt{3} + 4)$ 2
- c) Write the exact value of $\operatorname{cosec} 60^\circ$. 1
- d) Simplify $\frac{x-3}{x^2-4x+3}$ 1
- e) Find the value of x , correct to 1 decimal place. 2

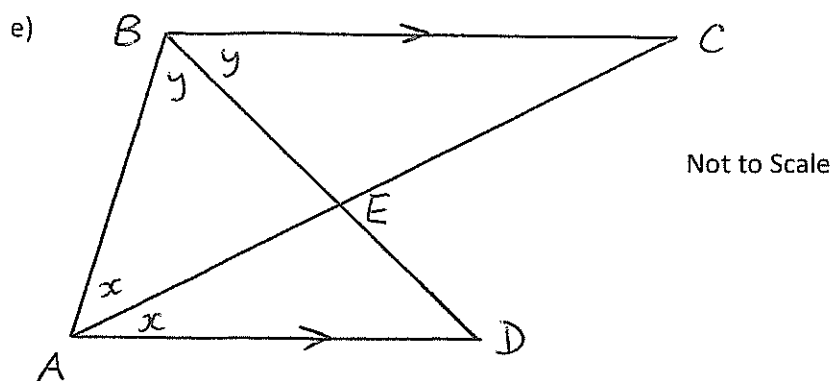
Not to Scale



- f) Solve $(x + 1)^2 = 5$, giving answers correct to 1 decimal place. 2

QUESTION 12 (9 marks) [Start a new page.](#)

- a) Solve $|3x - 6| < 12$ 2
- b) Find θ to the nearest degree if $\cos \theta = 0.4$ and $0^\circ \leq \theta \leq 360^\circ$. 1
- c) Fully simplify $\frac{a+b}{\frac{1}{a} + \frac{1}{b}}$ 2
- d) Find derivatives of: i) $y = 3x^2 - 4 + 7x$ 1
- ii) $f(x) = \frac{4}{x^2}$ 1



$AD \parallel BC$. AC and BD intersect at E . $\angle BAD$ and $\angle ABC$ are bisected as shown.

Prove that $\angle BEA = 90^\circ$.

2

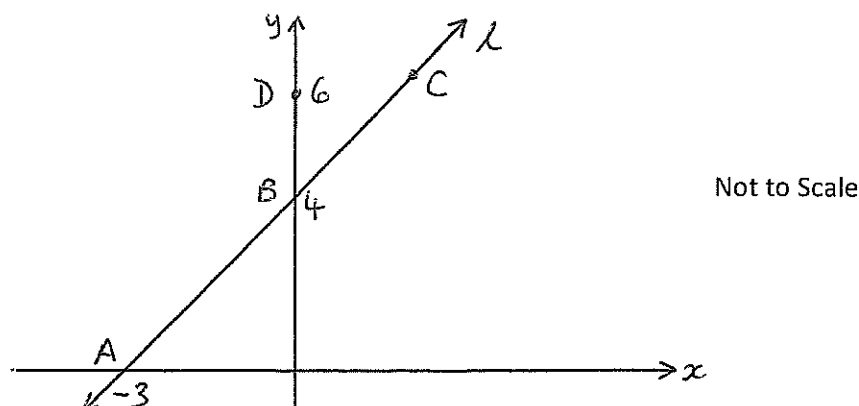
QUESTION 13 (9 marks) Start a new page.

- a) Factorise $x^3 - 27$. 1
- b) Simplify $\sin \theta (1 + \cot^2 \theta) \tan \theta$. 2
- c) Differentiate i) $(x^2 + 5)^4$ 1
- ii) $x\sqrt{x}$ 1
- d) Find the gradient of the curve $y = \frac{2x}{x+3}$ when $x = -2$. 2
- e) The solutions of a quadratic equation are $x = \frac{1 \pm \sqrt{5}}{2}$. Write a quadratic equation with these solutions. 2

QUESTION 14 (9marks) Start a new page.

- a) Find the coordinates of the vertex of the parabola $y = (x + 3)^2 + 4$. 1
- b) Given $f(x) = x^2 + \frac{x}{2}$, evaluate $f(2) + f'(2)$. 2
- c) Solve for θ , given $0^\circ \leq \theta \leq 360^\circ$:
- i) $(\sin \theta + 1)(\cos \theta - 1) = 0$ 2
- ii) $3\tan^2 \theta - 1 = 0$ 2
- d) If $\cos \theta = -\frac{2}{3}$ and $\sin \theta > 0$, find the exact value of $\tan \theta$. 1
- e) Fully factorise $x^2 + 8x + 16 - y^2$. 1

QUESTION 15 (9 marks) [Start a new page.](#)

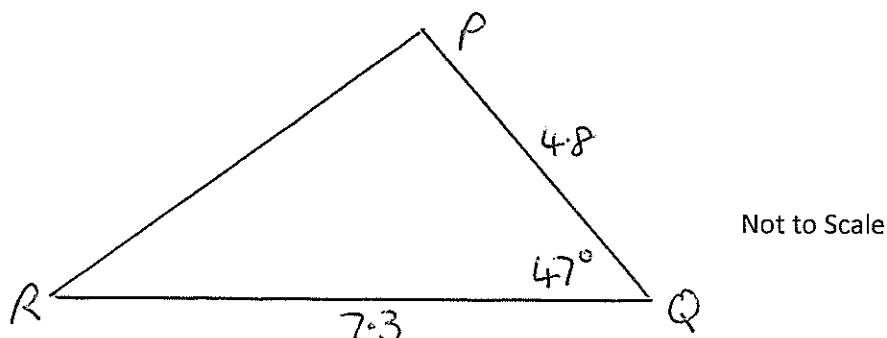


In the diagram above, line l cuts the x axis at $A(-3,0)$ and the y axis at $B(0,4)$. D has coordinates $(0,6)$ and point C is on l .

- Find the gradient of line l . 1
- Show that line l has equation $4x - 3y + 12 = 0$. 1
- B is the midpoint of AC . Find the coordinates of C . 1
- Find the perpendicular distance from D to the line l . 1
- Find the area of $\triangle BDC$. 2
- Find the equation of the perpendicular bisector of AB . Leave your answer in general form. 3

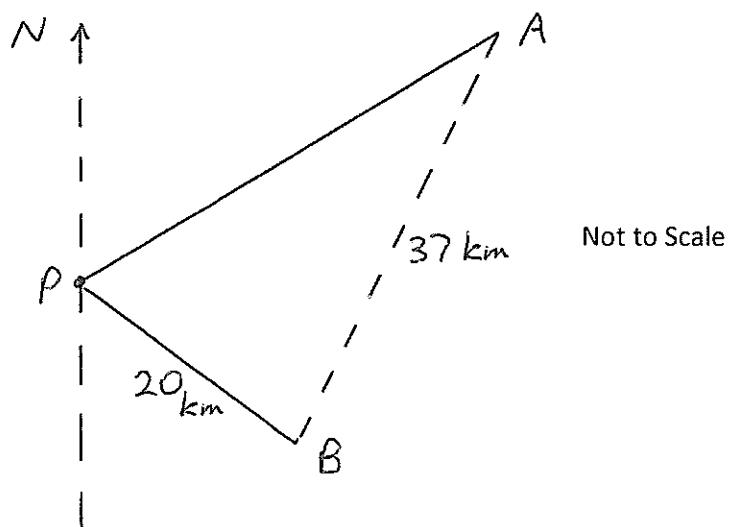
QUESTION 16 (9 marks) [Start a new page.](#)

- Find $\frac{d}{dr} \left(\frac{4}{3} \pi r^3 \right)$ 1
- Find the point(s) on the curve $y = x^3 - 3x^2 + 3x$ where the tangent is horizontal. 2
-



- Find the area of $\triangle PQR$. 1
- Find the length of RP , correct to 1 decimal place. 2

d)



Ship A leaves port P and sails on a compass bearing of $N50^\circ E$. Ship B also leaves port P and sails 20 km on a compass bearing of $S55^\circ E$. The two ships are now 37 km apart.

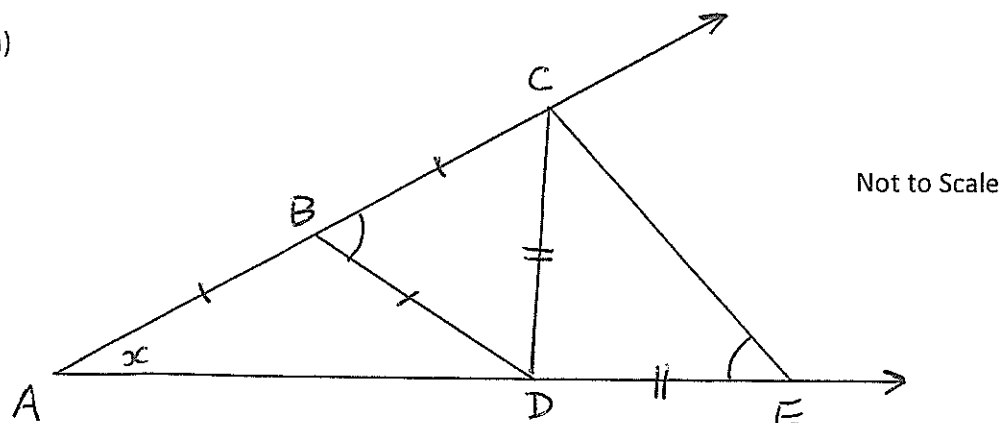
- i) Find $\angle APB$. 1
- ii) Find $\angle PAB$ to the nearest minute. 2

QUESTION 17 (9marks) Start a new page.

- a) The interior angle of a regular polygon is 165° . How many sides does the polygon have? 1
- b) Find the centre and radius of the circle $x^2 + y^2 - 4y - 1 = 0$. 2
- c) i) On the same axes, neatly sketch the functions $y = \frac{1}{x+2}$ and $y - x = 2$. Use a ruler, label any asymptotes and all x and y intercepts. 2
- ii) Find the points of intersection of the two graphs. Show working. 2
- iii) Find the equation of the normal to the curve $y = \frac{1}{x+2}$ at the point where it crosses the y axis. 2

QUESTION 18 (9 marks) Start a new page.

a)



Rays AC and AE enclose isosceles triangles ABD, BCD and CDE as shown above.

i) If $\angle A = x$, find $\angle CBD$ in terms of x , giving reasons.

2

ii) Hence, find the size of $\angle DEC$. Reasons are not required.

1

b) Simplify $\lim_{h \rightarrow 0} \frac{3(x+h)^2 - 3x^2}{h}$. Show full working.

2

c) Prove that $\sec \theta - \sin \theta \tan \theta = \cos \theta$.

2

d) Differentiate $y = x^2(x^3 - 1)^4$. Leave your answer in fully factored form.

2

END OF TEST

SOLUTIONS Yr 11 ~~Maths~~ Unit

1. D 2. B 3. C 4. A 5. A 6. B
 7. D 8. B 9. A 10. D

(11) a) 9.1

b) $6 + 8\sqrt{3} - \sqrt{3} - 4 \leftarrow \textcircled{1}$

$= 2 + 7\sqrt{3} \leftarrow \textcircled{1}$

c) $\frac{2}{\sqrt{3}}$

d)

$\frac{1}{x-1}$

e) $\cos 67^\circ = \frac{4.6}{x} \leftarrow \textcircled{1}$

$x = \frac{4.6}{\cos 67^\circ} \leftarrow \textcircled{1}$
 $= 11.8 \text{ (1 dec.)}$

f) $x+1 = \pm\sqrt{5} \leftarrow \textcircled{1}$

$x = -1 \pm \sqrt{5} \leftarrow \textcircled{1}$
 $= 1.2 \text{ or } -3.2 \leftarrow \textcircled{1}$
 (1 dec.)

(12) a) $3x-6 < 12 \text{ or } -(3x-6) < 12$

$3x < 18$

$-3x+6 < 12$

$x < 6$

$-3x < 6$

must have

$x > -2$

$\therefore$ solution is $-2 < x < 6$

b) $\theta = 66^\circ \text{ or } 29.4^\circ$

need both

c) $\frac{a+b}{\frac{b+a}{ab}} = \frac{a+b}{\frac{b+a}{ab}} \times \frac{ab}{ab}$

$= ab \leftarrow \textcircled{1}$

d) i) $y' = 6x+7$

ii) $f(x) = 4x^{-2}$

$f'(x) = -8x^{-3} \leftarrow \textcircled{1}$

or $-\frac{8}{x^3}$

$\textcircled{1}$

e) $2x+2y = 180^\circ$ (co-interior angles parallel lines)

$\therefore x+y = 90^\circ$

$\therefore \angle BEA = 90^\circ$ (angle sum $\triangle BEA$) $\textcircled{1}$

(13) a) $(x-3)(x^2+3x+9)$

b) $\sin \theta \times \operatorname{cosec}^2 \theta \times \tan \theta$

$= \cancel{\sin \theta} \times \frac{1}{\cancel{\sin^2 \theta}} \times \frac{\cancel{\sin \theta}}{\cos \theta} \textcircled{1}$

$= \frac{1}{\cos \theta} = \sec \theta \textcircled{1}$

c) i) $y' = 4(x^2+5)^3 \times 2x$
 $= 8x(x^2+5)^3$

ii) $y = x^{\frac{3}{2}}$

$y' = \frac{3}{2} x^{\frac{1}{2}}$

or $\frac{3\sqrt{x}}{2}$

$$\textcircled{13} d) y' = \frac{2(x+3) - 1(2x)}{(x+3)^2}$$

$$= \frac{6}{(x+3)^2} \leftarrow \textcircled{1}$$

When $x = -2$, $M_T = 6$ $\leftarrow \textcircled{1}$

e) $a = 1, b = -1, c = -1$

$$\therefore x^2 - x - 1 = 0 \leftarrow \text{must have } \textcircled{1}$$

$\textcircled{14} a) V(-3, 4)$

b) $f(2) = 5$

$$f'(x) = 2x + \frac{1}{2}$$

$$\therefore f'(2) = 4\frac{1}{2} \leftarrow \textcircled{1}$$

$$\therefore f(2) + f'(2) = 9\frac{1}{2} \leftarrow \textcircled{1}$$

c) i) $\sin \theta = -1$ or $\cos \theta = 1$

$$\therefore \theta = 270^\circ, 0^\circ, 360^\circ$$

ii)

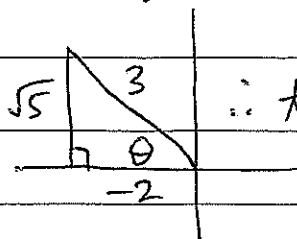
$$\tan^2 \theta = \frac{1}{3} \leftarrow \textcircled{1}$$

$$\therefore \tan \theta = \pm \frac{1}{\sqrt{3}}$$

$$\therefore \theta = 30^\circ, 150^\circ, 210^\circ, 330^\circ$$

all 4 $\textcircled{1}$

d) 2nd quadrant



$$\therefore \tan \theta = -\frac{\sqrt{5}}{2}$$

e) $(x+4)^2 - y^2$

$$= (x+4+y)(x+4-y)$$

$\textcircled{15} a) M_x = \frac{4}{3}$

b) use $B(0, 4)$

$$\therefore y - 4 = \frac{4}{3}(x - 0)$$

$$\therefore 3y - 12 = 4x$$

$$\therefore 4x - 3y + 12 = 0 \text{ as reqd.}$$

c) $C(3, 8)$

d) $D(0, 6)$ and $4x - 3y + 12 = 0$

$$\therefore \text{p.d.} = |0 - 18 + 12|$$

$$\sqrt{4^2 + 3^2}$$

$$= \frac{6}{5} \leftarrow \textcircled{1}$$

e)

$$B(0, 4) C(3, 8) \Rightarrow BC = \sqrt{9 + 16}$$

$$= 5 \leftarrow \textcircled{1}$$

$$\therefore \text{area } \triangle BDC = \frac{1}{2} \times 5 \times \frac{6}{5}$$

$$= 3 \text{ u}^2 \leftarrow \textcircled{1}$$

f) $MP(AB) = (-1\frac{1}{2}, 2) \leftarrow \textcircled{1}$

$$M_{AB} = \frac{4}{3} \Rightarrow M_{\perp} = -\frac{3}{4} \leftarrow \textcircled{1}$$

$$\therefore \text{eqn is } y - 2 = -\frac{3}{4}(x + 1\frac{1}{2})$$

$$4y - 8 = -3x - 4\frac{1}{2}$$

$$\therefore 6x + 8y - 7 = 0$$

$\leftarrow \textcircled{1}$

(16) a) $4\pi r^2$

b) $\frac{dy}{dx} = 0$

$\therefore 3x^2 - 6x + 3 = 0$
 $\therefore 3(x^2 - 2x + 1) = 0$ ①

$3(x-1)^2 = 0$ ①

$\therefore x = 1 \Rightarrow (1, 1)$ on curve

c) i) $A = \frac{1}{2} \times 7 \cdot 3 \times 4 \cdot 8 \times \sin 47^\circ$
 $\div 12.81 \text{ cm}^2$ ①

ii) $RP^2 = 7 \cdot 3^2 + 4 \cdot 8^2 - 2 \times 7 \cdot 3 \times 4 \cdot 8 \times \cos 47^\circ$
 $= 28.53555$

$\therefore RP \div 5.3$ (1 dec.)

①

d) i) 75°

ii) $\frac{20}{\sin A} = \frac{37}{\sin 75^\circ}$ ①

$\therefore \sin A = \frac{20 \sin 75^\circ}{37}$

$= 0.522 \dots$ ①

$\therefore \angle PAB = 31^\circ 28'$

(17) a) Ext. angle $= 15^\circ$

$\therefore \text{no. of sides} = \frac{360}{15}$

$= 24 \text{ sides.}$

b) $x^2 + y^2 - 4y + 4 = 1 + 4$

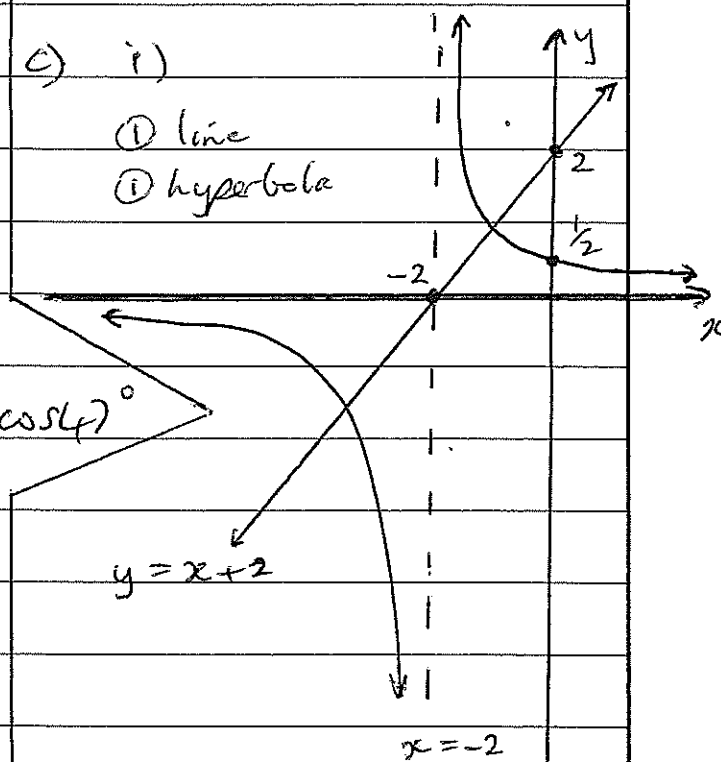
$x^2 + (y-2)^2 = 5$ ①

$\therefore$ centre is $(0, 2)$
 radius is $\sqrt{5} \text{ u}$ ①

c) i)

① line

① hyperbola



ii) $\frac{1}{x+2} = x+2$
 $(x+2)^2 = 1$
 $x^2 + 4x + 3 = 0$ ①

$(x+3)(x+1) = 0$

$\therefore x = -3, -1$

$\therefore$ pts. of intersection are $(-3, -1)$ and $(-1, 1)$

①

(17) c) iii) $y = (x+2)^{-1}$
 $y' = -1(x+2)^{-2} \times 1$

$$= \frac{-1}{(x+2)^2}$$

When $x = 0$, $y' = m_T = -\frac{1}{4}$

①

$\therefore m_N = 4$ and use $(0, \frac{1}{2})$

$\therefore$ eqn. of normal is

$$y - \frac{1}{2} = 4(x - 0)$$

$\therefore y = 4x + \frac{1}{2}$ ← ①

(or $8x - 2y + 1 = 0$)

(18) a) i) $\angle BDA = x$ (base angles ^① isosceles $\triangle ABD$)

$\therefore \angle CBD = 2x$ (exterior angle $\triangle ABD$) ^①

ii) 45°

b) $= \lim_{h \rightarrow 0} \frac{3(x^2 + 2xh + h^2) - 3x^2}{h}$

$$= \lim_{h \rightarrow 0} \frac{\cancel{3x^2} + 6xh + 3h^2 - \cancel{3x^2}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{K}(6x + 3h)}{\cancel{K}}$$

$= 6x$ ① method

① answer

c) LHS = $\frac{1}{\cos \theta} - \sin \theta \times \frac{\sin \theta}{\cos \theta}$

$$= \frac{1 - \sin^2 \theta}{\cos \theta} \quad \text{①}$$

$$= \frac{\cos^2 \theta}{\cos \theta}$$

$$= \cos \theta$$

= RHS. ①

d) $y' = 2x(x^3 - 1)^4$ ①
 $+ 4(x^3 - 1)^3 \times 3x^2 \times x^2$

$$= 2x(x^3 - 1)^4 + 12x^4(x^3 - 1)^3$$

$$= 2x(x^3 - 1)^3 [x^3 - 1 + 6x^3]$$

$$= 2x(x^3 - 1)^3 (7x^3 - 1) \quad \text{①}$$